Mathematical Analysis of Free-Rider Problems in Academic Settings

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1. INTRODUCTION

The concept of free-riding was first introduced and examined by Mancur Olson in his book The Logic of Collective Action. In this work, Olson highlighted the surprising tendency within small groups with shared interests for a minority to "exploit" the majority—a phenomenon he identified as the free-rider tendency (Golembiewski & Olson, 1966).

In this study, we define a free-rider as a group member whose contribution falls below the group's average output. To investigate the free-riding behavior, we establish a specific scenario of team collaboration within academic report projects on campus. It employs the game theory, mainly about the dominant strategy, to explore the reasons why free-rider problem happens and its solutions. Specifically, the research scenario involves a team of two members, Person A and Person B, working together on a time-limited learning task, where they must complete a project within a set timeframe with a final output in the form of a research report.

2. METHODOLOGY

2.1. Notation and Symbol

Table 1. Notation and Symbol

Symbol	Definition	Unit
e_A, e_B	e_A represents Person A's efficiency; e_B represents Person B's efficiency; $e_A \geq 0$, $e_B \geq 0$	100 words/hour
m	Working time duration for Person A, $m \in \{Z \mid 0 \leq m \leq x\}$	hour
n	Working time duration for Person B, $n \in \{Z \mid 0 \leq n \leq x\}$	hour
x	Maximum working hours, $x \in Z$	hour
$a_{i,j}$	Perceived payoff for Person A when Person A works m hours and Member B works n hours; i=m+1, j=n+1	score
$b_{i,j}$	Perceived payoff for Person B when Person A works m hours and Person B works n hours; i=m+1, j=n+1	score
A	Matrix of perceived payoff for Person A, $A=(a_{i,j})_{(x+1)*(x+1)}$, with i and j as row and column indices, respectively; i=m+1, j=n+1	\
B	Matrix of perceived payoff for Person B, $B=(b_{i,j})_{(x+1)*(x+1)}$, with i and j as row and column indices, respectively; i=m+1, j=n+1	\
α	Working time for Person A's dominant strategy, $\alpha \in \{Z \mid 0 \leq \alpha \leq x\}$	hour
β	Working time for Person B's dominant strategy, $\beta \in \{Z \mid 0 \leq \beta \leq x\}$	hour
k	Free-rider index for Person A at dominant strategy	\

	equilibrium	
$C_A(m)$, $C_B(n)$	Perceived time cost functions for Person A and B, respectively	100 words
d_{effi}	Efficiency difference between Person A and Person B	100 words/hour
d_{cost}	Perceived time cost function's difference between Person A and Person B	\
$v_{i,j}$	Perceived total group output when Person A works m hours and Person B works n hours; i=m+1, j=n+1	100 words
γ	Perceived total group output at dominant strategy equilibrium	100 words
V	Matrix of perceived total group output, $V=(v_{i,j})_{(x+1)*(x+1)}$, with i and j as row and column indices, respectively; i=m+1, j=n+1	\

In this study, e_A and e_B represent the efficiency of Person A and Person B, respectively. Here, the efficiency is defined as the output per unit time, with "hours" as the time unit and the effective word count as the measure of output. Since the final output of this project is a joint research report, the key is to maximize the effective content of the report, so we choose "effective word count" as our output unit, with "100 words" representing one "word-unit." At the same time, e_A, $e_B \geq 0$ assuming that none of Person A and Person B contributes negatively. $d_{effi}=e_A-e_B$, when $d_{effi}=0$, Person A and B's efficiencies are equal; when $d_{effi}>0$, Person A has a higher efficiency than Person B; similarly, when $d_{effi}<0$, Person B has a higher efficiency than Person A.

This study is based on a scenario which involves with a time-limited task, so Person A and Person B must complete the report within x hours. Consequently, both Person A's working time, m, and Person B's working time, n, are restricted to values from 0 to x (assuming integer values). Also, the perceived benefits (payoffs) $a_{i,j}$ and $b_{i,j}$ represent the subjective payoff that Person A and Person B assign to their combined works when Person A works for (i-1) hours and Person B works for (j-1) hours. Given that $m,n \in \{Z|0 \leq m,n \leq x\}$, there are (x+1) strategy options for each person and, hence, $(x+1)^2$ distinct combinations of working hours for both. For simplicity, we index the perceived benefits (payoffs) $a_{i,j}$ and $b_{i,j}$ for each combination of Person's A and Person B's working time from (1,1) to (x+1,x+1).

Matrix A and B list the perceived benefits (payoffs) of each possible strategy combination for Person A and Person B respectively. The values of α and β represents the working time chosen by Person A and Person B when they adopt their dominant strategies. If multiple dominant strategies exist for either person, α and β are defined as the minimum working time among these strategies. This is because, with identical expected payoffs, players tend to prefer strategies that require less time investment.

The free-riding index k for Person A at dominant strategy equilibrium is defined as:

$$k = \frac{\alpha \cdot e_A}{\gamma} - \frac{1}{2}$$

This expression indicates the difference between the ratio of Person A's contribution to the total output

and 1/2, which serves as a measure for determining free-riding. When k>0, Person A contributes a greater share than half of the total output and, therefore, does not free-ride. Conversely, when k<0, Person A's contribution is less than half of the total output, suggesting free-riding. The degree of free-riding is shown by |k|. If |k|=0, both people contribute equally, and no free-riding happens.

The perceived time cost function $C_A(m)$ and $C_B(n)$ (also in unit of "100 words") represent Person A and Person B's perceived time costs of their working time. These functions are non-linear and increase with additional time spent, reflecting a convex cost perception where early contributions are felt less costly compared to extended hours, which carry a higher perceived cost. To compare these perceived time cost of Person A and Person B, we define d_{cost}=In(f)-In(g), where f and g are scaling parameters of the perceived time cost functions. When $d_{cost}=0$, Person A and Person B perceive time costs equally; when $d_{cost}>0$, Person A perceives a higher cost; when $d_{cost}<0$, Person B perceives a higher cost.

2.2. Basic Assumption

To simplify the modeling and analysis, we make following assumptions:

a. **Rationality:** Each participant aims to maximize their perceived benefit with least possible time investment and acts independently, without bias or influences by other personal factors.
b. **Perceived and Actual Output Equality:** Participant's perceived total output is assumed equal to actual output, with a conversion where 1 point is awarded to 1 word-unit of the report.
c. **Fixed Working Time:** Each participant's available work time has an upper limit, fixing the maximum task completion time.
d. **Nonlinear Time Costs:** Perceived time costs increase non-linearly with time, so incremental time feels progressively more costly.
e. **Linear Group Output:** It is assumed that the collective group output is linearly related to the input of each member, which simplifies the process of converting individual contributions into collective payoffs.
f. **Independent Cooperation:** Each session of cooperation is independent, so individual reputation and cumulative effects are not considered.

2.3. Modeling

As previously mentioned, this study focuses on a specific scenario which will be discussed in detail in this section. Based on the scenario, we will develop a model for it.

In the scenario we study, Person A and Person B work together on a time-limited task, with the goal of producing results within a limited period in the form of a research report. The available working time for both individuals is fixed, as this imitates real-life situations. "Available working time" is defined as the duration from the start of collaboration to the final task deadline, a concept we frequently encounter in our daily life, such as an 8-hour workday or a 2-hour exam duration. The output in this scenario is graded and shared together, with no division of payoffs or outputs.

This study investigates the strategy combinations for Person A and Person B under various circumstances by changing their efficiency level, e_A and e_B, and their perceived time cost function, $C_A(m)$ and $C_B(n)$. The objective is to identify a dominant strategy for their working time, where neither can achieve a higher payoff by changing their strategy regardless of other's choice. The concept of

"Person A's free-riding index," k, is used to examine the occurrence trends of free-riding under different situations. The following definitions are established:

$$v_{i,j} = e_A \cdot m + e_B \cdot n \quad (1)$$

, where $v_{i,j}$ represents the perceived total payoff by the team when Person A works for m hours and Person B works for n hours, with e_A and e_B as efficiency of Person A and Person B, respectively. Because the model assumes that the perceived total payoff is equal to the actual total payoff, the perceived total payoff can be determined by summing the outputs of both individuals.

$$a_{i,j} = v_{i,j} - C_A(m) \quad (2)$$

$$b_{i,j} = v_{i,j} - C_B(m) \quad (3)$$

Here, $a_{i,j}$ and $b_{i,j}$ represents the individual perceived payoff for both Person A and Person B, respectively, when Person A works for m hours and Person B works for n hours. $C_A(m)$, $C_B(n)$ are the perceived time cost function, respectively. Based on our previous assumptions and descriptions, $C_A(m)$, $C_B(n)$ are given by $C_A(m)=f^m-1$, $C_B(n)=g^n-1$, where f>1 and g>1. Since payoff equals output minus cost, the perceived payoff for each team member is calculated as perceived total payoff minus perceived time cost.

2.4. General Solution for Dominant Strategy

Dominant strategy in game theory refers to a strategy that always yields a better payoff than any other strategy a player may choose. It is a solution for the game and guarantees a higher payoff for the player ("Cybersecurity and Applied Mathematics," 2016).

Step 1: Determine the maximum available working time x, the efficiencies e_A and e_B for Person A and Person B, and the perceived time cost function $C_A(m)$ and $C_B(n)$.

Step 2: To analyze the free-riding behavior trends, let m, n represent the working hours for Person A and Person B as integer values from 0 to x, with measured in hours. Using equation (1), (2) and (3), calculate the perceived total output $v_{i,j}$, Person A's perceived output $a_{i,j}$ and Person B's perceived output $b_{i,j}$, for each possible combination of working time, and construct matrices V, A, B where values in matrices depends on indices i and j.

Step 3: Based on the matrices, find dominant strategy equilibrium to determine the optimal time allocation strategy that maximizes the perceived total output and the dominant strategies for each individual.

Step 3.1. To determine Person A's optional strategy, solve:

$$\max_{\{1 \le i \le x+1\}} \left(\min_{\{1 \le j \le x+1\}} a_{i,j} \right) = a_{i^*,j^*}$$

Here, the working hour (i^*-1) is the dominant strategy of Person A. This is determined by evaluating the minimum payoff $a_{i,j}$ for each fixed working time m hour that Person A may choose, and then selecting the working time that maximizes these minimum payoffs.

The logic is as follows: when Person A works for a fixed duration m hour, each additional hour that Person B works increases Person A's perceived payoff by e_B. Therefore, the minimum payoff for Person A at any fixed m hour occurs when Person B works 0 hours (n=0). Since Person A's perceived payoff

increases linearly with respect to Person B's working hours, choosing strategies that maximize the minimum payoff at n=0 ensures that Person A selects a strategy that guarantees the highest achievable payoff, regardless of Person B's choice. Thus, the working hour (i^*-1) is the dominant strategy for Person A.

Step 3.2. Similarly, the Person B's dominant strategy is given by:

$$\max_{\{1 \le j \le x+1\}} \left(\min_{\{1 \le i \le x+1\}} b_{i,j} \right) = b_{i^{**},j^{**}}$$

, where (j^{**} -1) is the working hour for Person B at dominant strategy equilibrium.

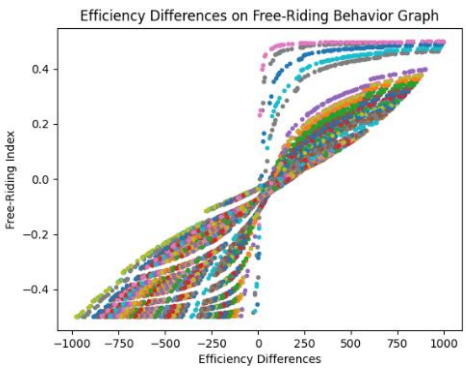
Step 3.3. Among the multiple dominant strategy working hour pairs, select the one where both players work the minimum amount of time. The solution of the game is given by this pair. Based on this selected pair, calculate the perceived total output γ and Person A's free-riding index k.

3. DISCUSSION

In this section, we conduct two experiments to investigate the impact of efficiency difference and perceived time cost function difference on free-riding behavior. We analyze and interpret the observed data from two experiments, which provides us implications on the free-riding behavior trends. Further, based on our findings, we conclude some valuable strategies for reducing free-riding behavior in more general collaborative settings.

3.1. Experiment 1: Impact of Efficiency Differences on Free-Riding Behavior

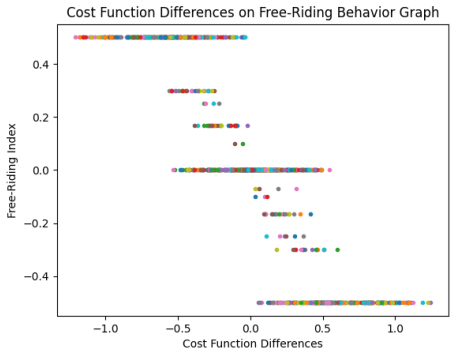
In the first experiment, we used our model to investigate how the difference between the efficiencies of Person A and Person B can influence the free-riding behaviors. In the experiment, given the fixed available working time (x=8) and perceived time cost functions of both individuals (f=g=e), we randomly generate 10,000 pairs of efficiencies of Person A and Person B to observe the relationship between the free-riding index k and the efficiency difference d_{effi}, where $d_{effi}=e_A-e_B$. As depicted in the scatter plot graph, the result suggests that a smaller efficiency difference between individuals is associated with a lower degree of free-riding behavior. This trend aligns with our intuition that when one team member is more efficient than another, the less efficient individual may be more inclined to reduce their contribution, relying instead on the efforts of the higher-performing teammate.



3.2. Experiment 2: Impact of Perceived Time Cost Function Differences on Free-Riding Behavior

In the second experiment, we examine how the difference between the perceived time cost function affects the free-riding behavior through our model. Given the fixed available working time (x=8) and efficiency of both individuals ($e_A=e_B=3$), we randomly generate 1,000 pairs of perceived time cost function, ($C_A(m)$,$C_B(n)$), for Person A and B. In the experiment, the difference between the perceived

time cost functions is measured by d_{cost}, where d_{cost}=In(f)-In(g). As depicted in the second scatter plot graph, the result implies that smaller differences in perceived time cost result in lower levels of free-riding behavior. In other words, when one team member perceives their time cost as much higher than the other, they may be less motivated to contribute proportionally, leading to free-riding.



3.3. Implications and More Applications

Based on the previous modeling, solution, and analysis, several conclusions can be drawn, which can be divided into three main aspects.

a) When team members are similar abilities and valuations, specifically the efficiencies and perceived time costs, the free-riding behavior is unlikely to occur.
b) When there exists significant difference in efficiencies among team members, the less efficient individuals tend to incline on others' contribution, becoming the free-rider.
c) When there exists significant difference in perceived time cost among team members, the individuals with high per unit perceived time cost are less-motivated to contribute on the teamwork, leading to free-riding behavior.

These findings have practical implications for team-based environments, such as corporate project teams, supply chain collaborations, and joint academic ventures. By forming teams in a strategic manner, comprising members with minimal differences in efficiency and cost perception, managers can create conditions that minimize the free-riding and enhance the collective productivity.

4. CONCLUSION

This study analyzes free-riding behavior in the academic settings through a game-theoretical model. Based on results of experiment 1 and 2, we can conclude that the free-riding behavior can be effectively reduced by selecting team members with comparable efficiencies and aligning perceptions of time cost. These model's conclusions and findings can provide us valuable insights on more general collaborative settings, such as cooperative tasks within corporations and supply chain management.

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